LP 4

distanta max. 1

CD \ AB	$\bar{A}\bar{B}$ (00)	$\bar{A}B$ (01)	AB (11)	$A\bar{B}$ (10)
$\bar{C}\bar{D}$ (00)	$\bar{A}\bar{B}\bar{C}\bar{D}$ #0 0000	$\bar{A}B\bar{C}\bar{D}$ #4 0100	$AB\bar{C}\bar{D}$ #12 1100	$A\bar{B}\bar{C}\bar{D}$ #8 1000
$\bar{C}D$ (01)	$\bar{A}\bar{B}\bar{C}D$ #1 0001	$\bar{A}BCD$ #5 0101	$ABCD$ #13 1101	$A\bar{B}CD$ #9 1001
CD (11)	$\bar{A}\bar{B}CD$ #3 0011	$\bar{A}BCD$ #7 0111	$ABCD$ #15 1111	$A\bar{B}CD$ #11 1011
$C\bar{D}$ (10)	$\bar{A}B\bar{C}D$ #2 0010	$\bar{A}B\bar{C}D$ #6 0110	$AB\bar{C}D$ #14 1110	$A\bar{B}\bar{C}D$ #10 1010

A - MSB

D - LSB

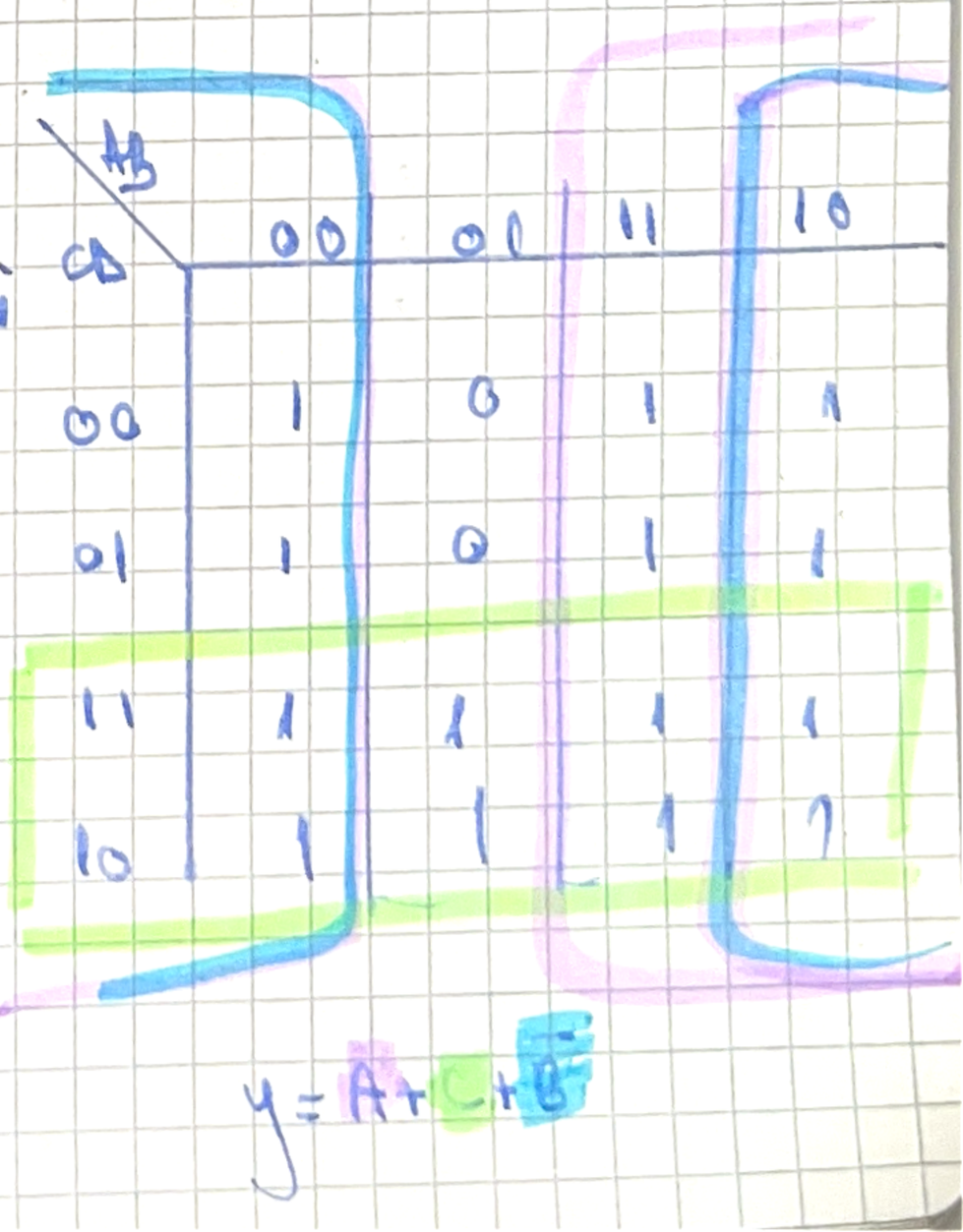
- $y : \Sigma_2^4 \rightarrow \Sigma$

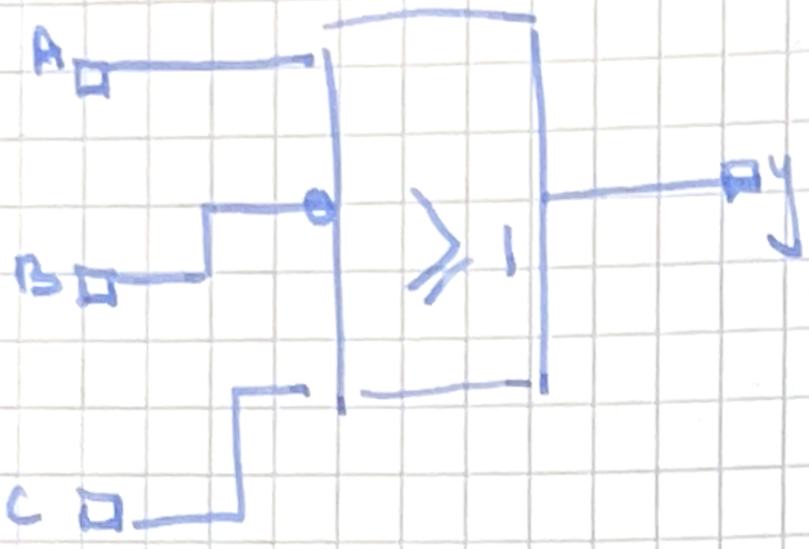
$$y : mc(0,1,2,3,6,7,8,9,10,11,12,13,14,15)$$

$$F_{CD} = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}BCD + ABCD + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D + \bar{A}\bar{B}\bar{C}D + \bar{A}BCD + AB\bar{C}\bar{D} + AB\bar{C}D + ABC\bar{D} + A\bar{B}CD + \bar{A}\bar{B}C\bar{D}$$

$$F_{CC} = M(4,5) = (A + \bar{B} + C + D)(A + \bar{B} + C + \bar{D})$$

A	B	C	D	y	
0	0	0	0	1	
0	0	0	1	1	
0	0	1	0	1	
0	0	1	1	1	
0	1	0	0	0	#4 y Maxterm
0	1	0	1	0	#5
0	1	1	0	1	
0	1	1	1	1	
1	0	0	0	1	
1	0	0	1	1	
1	0	1	0	1	
1	0	1	1	1	
1	1	0	0	1	
1	1	0	1	1	
1	1	1	0	1	
1	1	1	1	1	





- $y : \Sigma_2^4 \longrightarrow \mathcal{E}$

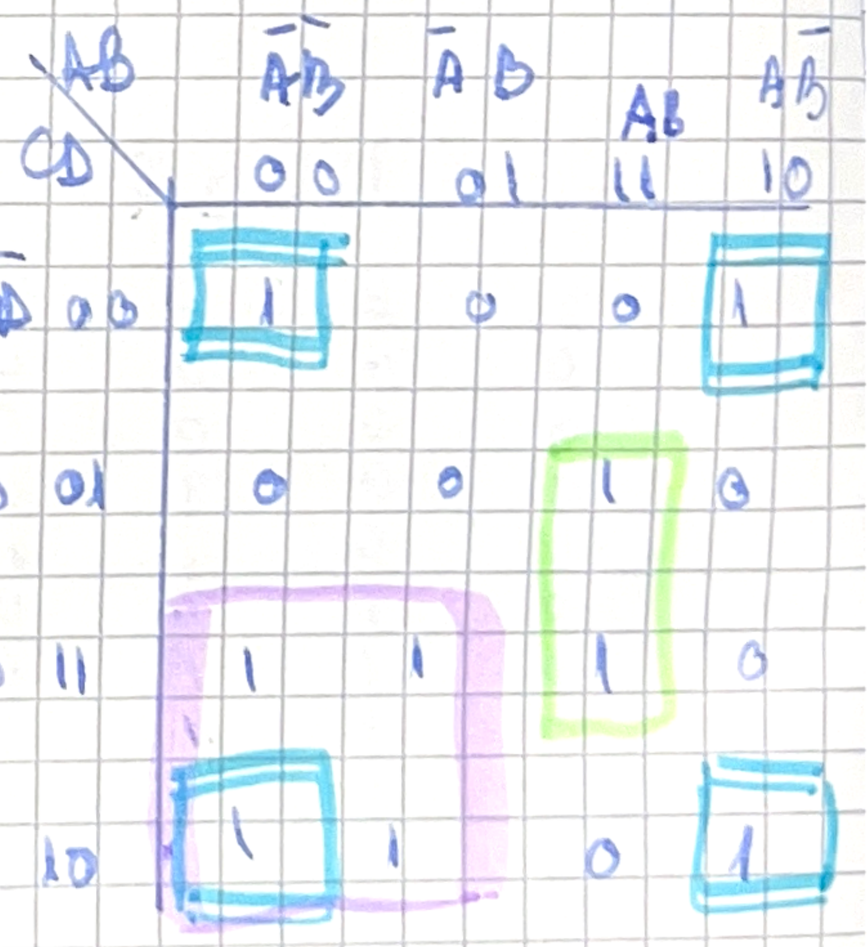
$$y = m(0,2,3,6,7,8,10,13,15)$$

$$FCD_y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}CD + A\bar{B}\bar{C}D + A\bar{B}CD + AB\bar{C}\bar{D} + A\bar{B}C\bar{D} + A\bar{B}CD + AB\bar{C}D + AB\bar{C}\bar{D}$$

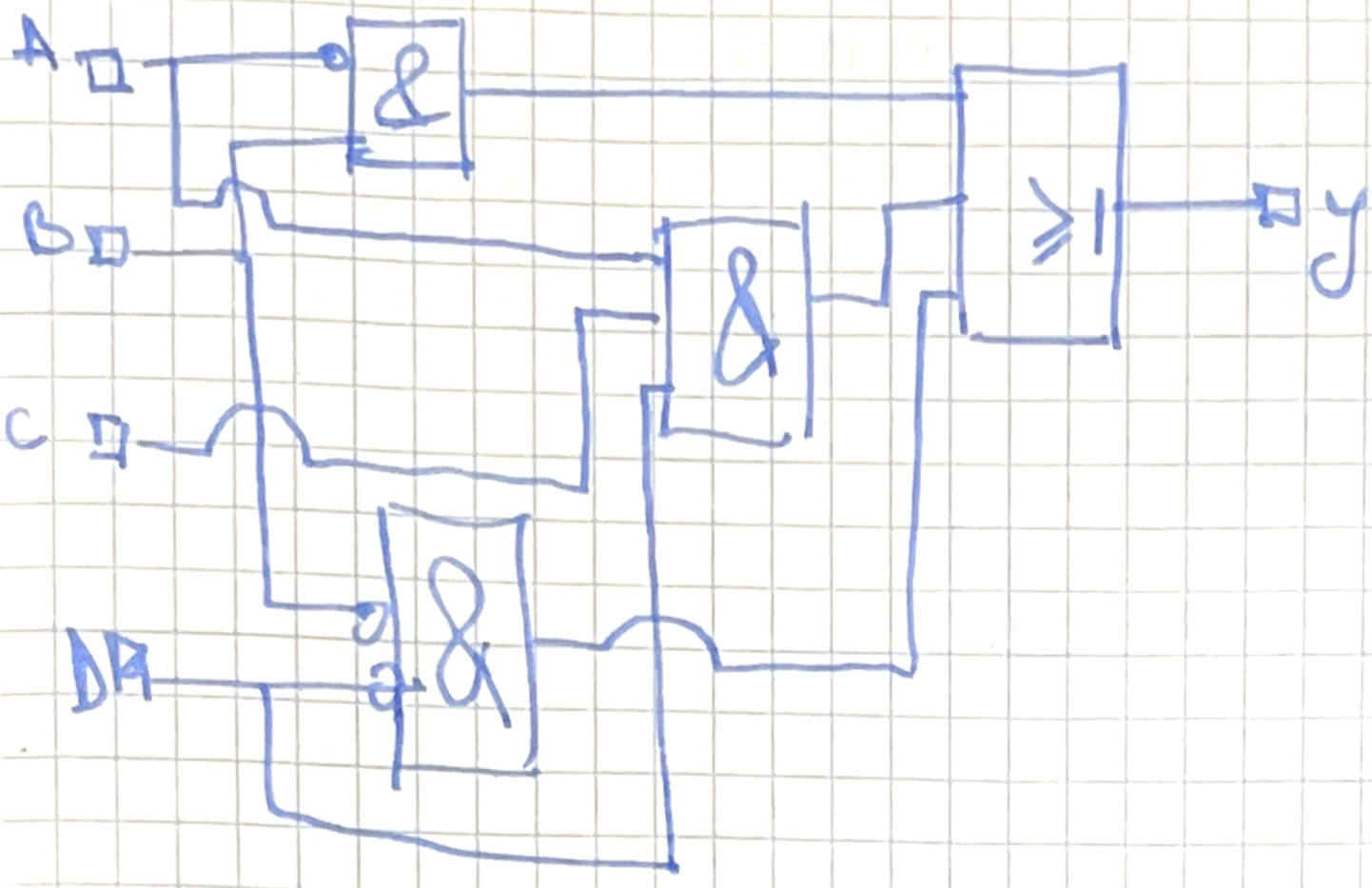
$$FCC = M(1,4,5,9,11,12,14)$$

$$= (A+B+C+\bar{D})(A+\bar{B}+C+D)(A+\bar{B}+C+\bar{D})(\bar{A}+B+C+\bar{D})(\bar{A}+B+\bar{C}+\bar{D})(\bar{A}+\bar{B}+C+D)$$

	A	b	C	D	y
0	0	0	0	0	1
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	1
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	



$$y = AC + \bar{B}\bar{D} + ABD$$



- $y : \Sigma_2^4 \to \Sigma$

$y = m(0, 5, 7, 8, 12, 13, 15)$

FCD $y = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}B\bar{C}D + \bar{A}BCD + A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} + ABCD + ABC\bar{D}$

FCC $y = M(1,2,3,4,6,9,10,11,14)$

$= (A+B+C+\bar{D})(A+B+\bar{C}+A)(A+B+\bar{C}+\bar{D})(A+\bar{B}+C+D)$
$(A+\bar{B}+\bar{C}+D)(\bar{A}+B+C+D)(\bar{A}+B+\bar{C}+D)(\bar{A}+D+\bar{C}+\bar{D})$
$(\bar{A}+\bar{B}+\bar{C}+D)$

	A	B	C	D	y
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	0
3	0	0	1	1	0
4	0	1	0	0	0
5	0	1	0	1	1
6	0	1	1	0	0
7	0	1	1	1	1
8	1	0	0	0	1
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	0
15	1	1	1	1	1

CD \ AB	$\bar{A}\bar{B}$ 00	$\bar{A}B$ 01	AB 11	$A\bar{B}$ 10
$\bar{C}\bar{D}$ 00	1	0	1	1
$\bar{C}D$ 01	0	1	1	0
CD 11	0	1	1	0
$C\bar{D}$ 10	0	0	0	0

$y = \bar{B}\bar{C}\bar{D} + BD + AB\bar{C}$

$\bar{A}\bar{B}$ $\bar{A}B$ $\bar{B}\bar{C}\bar{D}$

$\bar{A}\bar{B} \cdot AB \cdot \bar{C}D \cdot CD = BD$

$AB \cdot \bar{C}\bar{D} \cdot C\bar{D}$

$AB\bar{C}$

A
Si | Si | 0 | 1
Si | * | 1
0

Tema 1

10	numere	0-9
6	litere	10-15

$y: m(0,2,3,6,8,9,\overset{10}{A},\overset{11}{B},\overset{14}{E},\overset{15}{F})$

FCD y: $\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + A\bar{B}\bar{C}\bar{D} + A\bar{B}\bar{C}D$
$+ A\bar{B}CD + A\bar{B}C\bar{D} + ABC\bar{D} + ABCD$

FCC y: $M(1,4,5,7,\overset{12}{C},\overset{13}{D})$
$= (A+B+C+\bar{D})(A+\bar{B}+C+D)(A+\bar{B}+C+\bar{D})(A+\bar{B}+\bar{C}+D)$
$(\bar{A}+\bar{B}+C+D)(\bar{A}+\bar{B}+C+\bar{D})$

	A	B	C	D	y
#0	0	0	0	0	0
#1	0	0	0	1	0
#2	0	0	1	0	1
#3	0	0	1	1	1
#4	0	1	0	0	0
#5	0	1	0	1	0
#6	0	1	1	0	1
#7	0	1	1	1	0
#8	1	0	0	0	1
#9	1	0	0	1	1
A #10	1	0	1	0	1
B #11	1	0	1	1	1
C #12	1	1	0	0	0
D #13	1	1	0	1	0
E #14	1	1	1	0	1
F #15	1	1	1	1	1

distanta HAMMING

CD \ AB	$\bar{A}\bar{B}$ 00	$\bar{A}B$ 01	AB 11	$A\bar{B}$ 10
$\bar{C}\bar{D}$ 00	1	0	0	1
$\bar{C}D$ 01	0	0	0	1
CD 11	1	0	1	1
$C\bar{D}$ 10	1	0	1	1

PADURI CÂT MAI MARI!
16/8/4/2

$\bar{A}\bar{B}\bar{C}\bar{D} \cdot \bar{A}\bar{B}\bar{C}D \cdot \bar{A}\bar{B}C\bar{D} \cdot \bar{A}\bar{B}CD = \bar{B}\bar{D}$

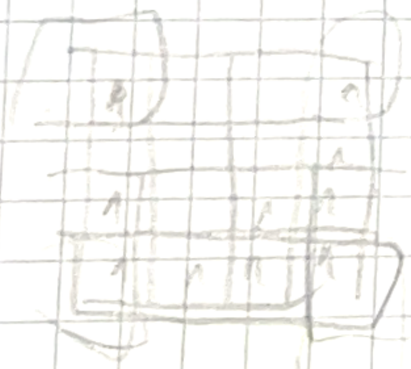
$\bar{A}\bar{B}C\bar{D} \cdot \bar{A}BC\bar{D} \cdot A\bar{B}C\bar{D} \cdot ABC\bar{D} = C\bar{D}$

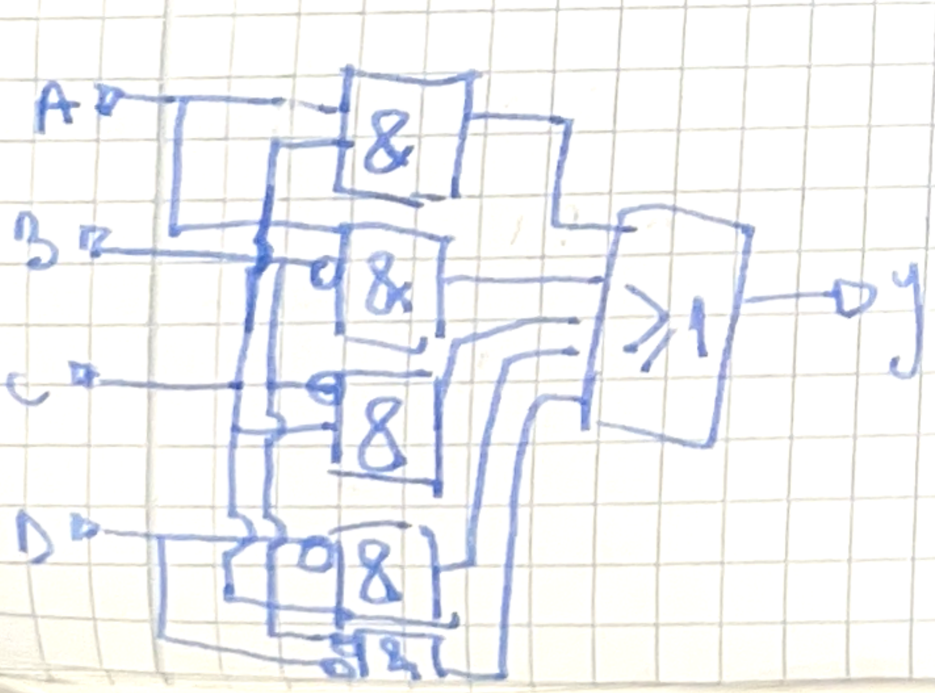
$A\bar{B}C\bar{D} \cdot A\bar{B}CD \cdot A\bar{B}C\bar{D} \cdot A\bar{B}CD = A\bar{C}$

$A\bar{B}\bar{C}\bar{D} \cdot A\bar{B}C\bar{D} = A\bar{B}\bar{C}$

$\bar{A}\bar{B}C\bar{D} \cdot \bar{A}\bar{B}CD = \bar{A}\bar{B}C$

$\bar{A}\bar{B}C\bar{D} \cdot \bar{A}\bar{B}CD$





$y: AC + A\bar{B}\bar{C} + \bar{A}\bar{B}C + C\bar{D} + \bar{B}\bar{D} = AC + A\bar{B} + \bar{B}C + C\bar{D} + \bar{B}\bar{D}$

$= A(C + \bar{B}\bar{C}) + C(\bar{D} + \bar{A}B) + \bar{B}\bar{D}$

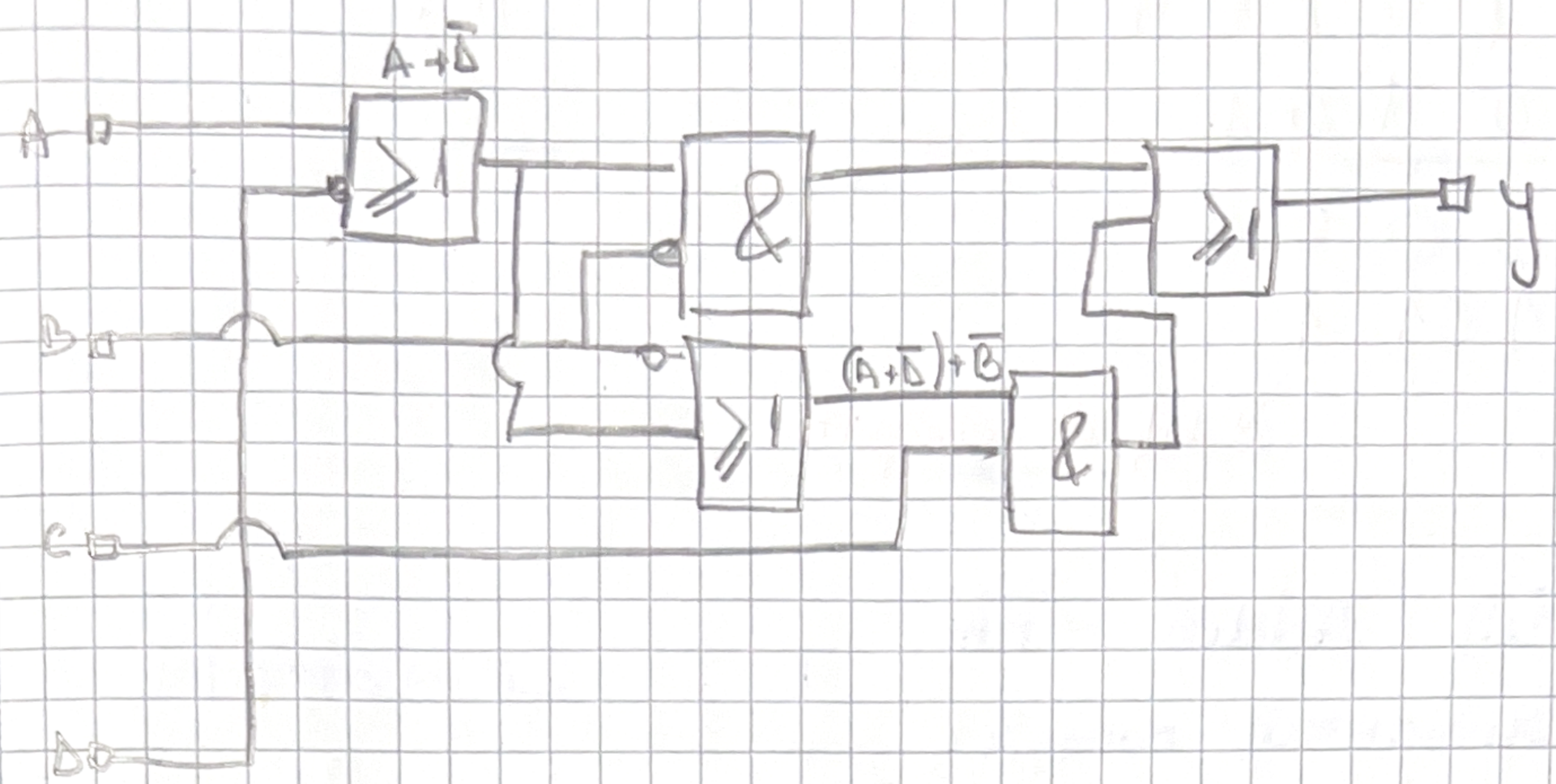
$= AC + A\bar{B} + \bar{B}C + C\bar{D} + \bar{B}D$

$B(A\bar{C} + \bar{A}C) = B(A+C)$

$A\bar{B}C + \bar{A}\bar{B}C = A\bar{B} + \bar{B}C = \bar{B}(A+C)$

$B(1$

LP5



Half Adder – HA

Semisumatorul

	A	B	S	C_0
#0	0	0	0	0
#1	0	1	1	0
#2	1	0	1	0
#3	1	1	0	1

C_0 - bit de transport
Carry - Out

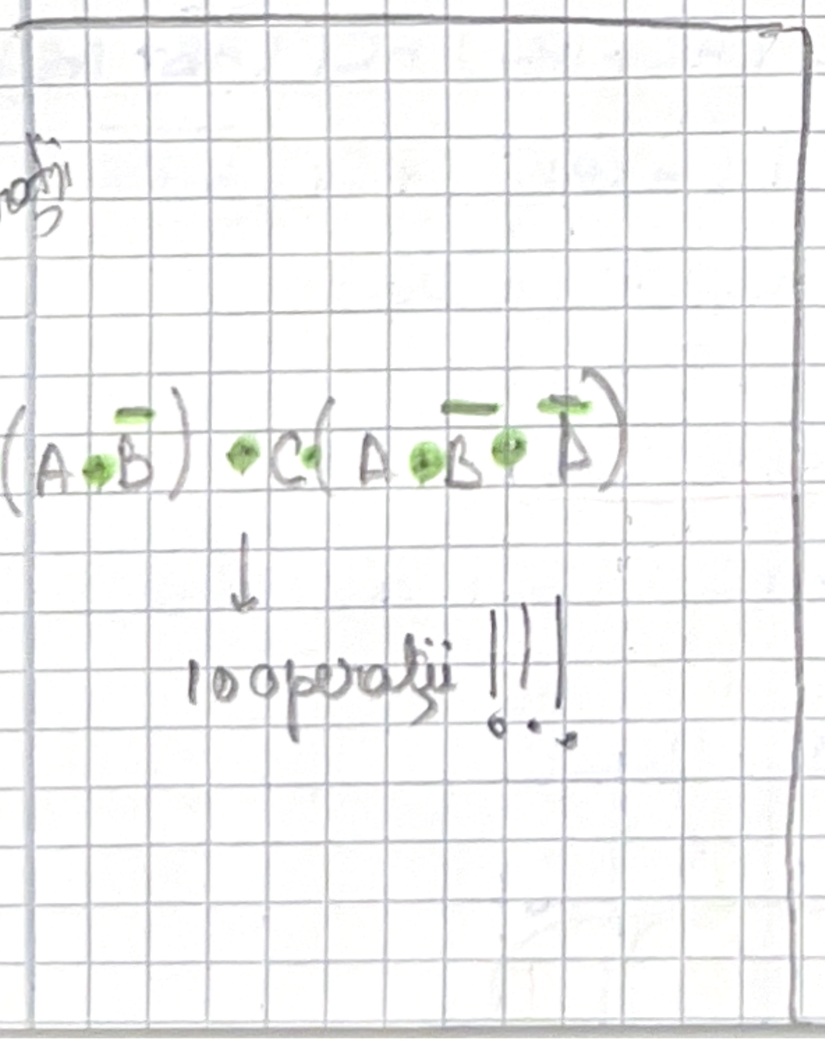
$\{C_0, S\} = \Sigma_2^2 \rightarrow \Sigma_2^2$

$S = \Sigma_2^2 \rightarrow \Sigma_2$

$C_0 = \Sigma_2^2 \rightarrow \Sigma_2$

$S = m(1,2) = A \oplus B$

$C_0 = m(3) = AB$

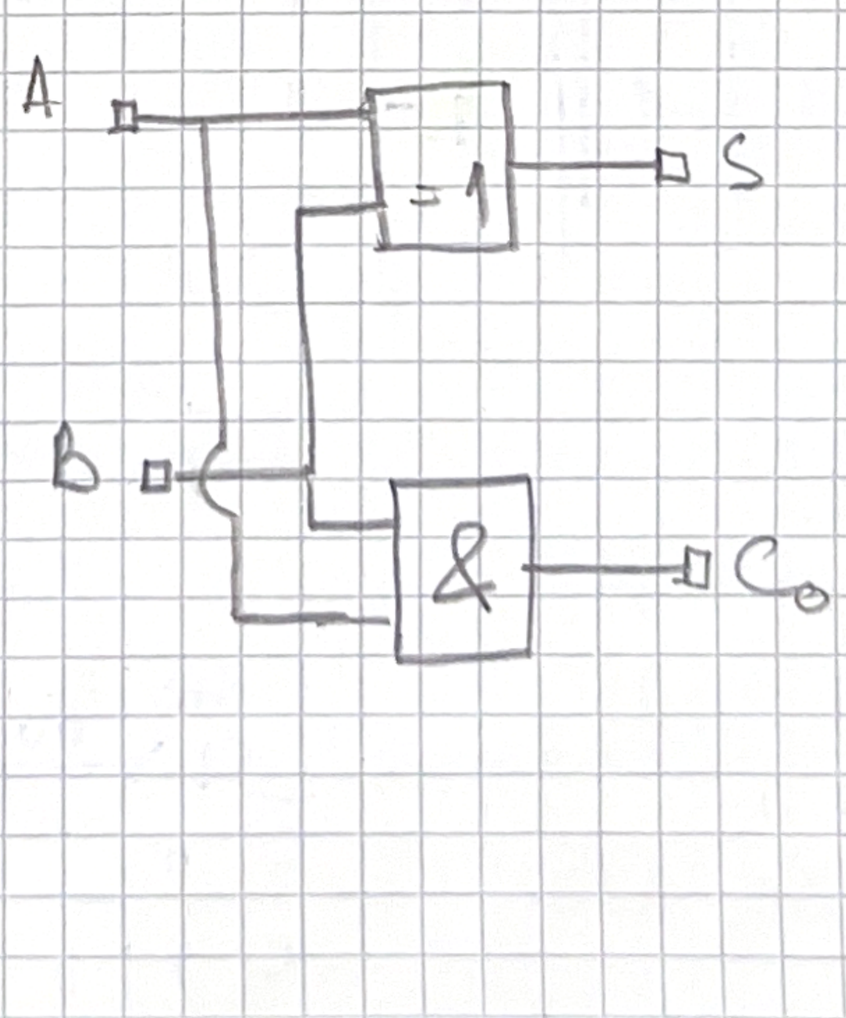


14 operații

$= \overline{\overline{B} \cdot (A + \overline{B})} + C(A + \overline{B} + \overline{D})$

↓

10 operații !!!



0 1 0 1 1 7

0 0 1 0 1

1 0 0 0 0

↑ ↑ ↑ ↑

↳ biți de transport

11 +

5

Full Adder - FA

Sumatorul complet

C_i - Carry in

$\{C_0, S\ y : \Sigma_2^3 \to \Sigma_2^2 = \begin{cases} S : \Sigma_2^3 \to \Sigma_2 \\ C_0 : \Sigma_2^3 \to \Sigma_2 \end{cases}$

Spațiul valorilor binare

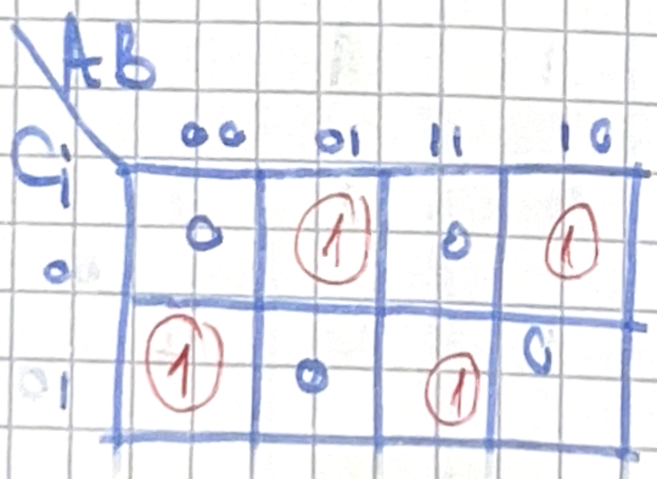
mapat în spațiul binar de 2 valori (binar)

$S = m(1,2,4,7)$

Sumă: $C_0 = B_0 = \bar{A}B + A\bar{B} = A \oplus B$

	A	B	C_i	S	C_0
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

A-B.



$2^3 = 8$ paduri

$F_{CD}: \bar{A}B\bar{C}_i + A\bar{B}\bar{C}_i + \bar{A}\bar{B}C_i + ABC_i =$

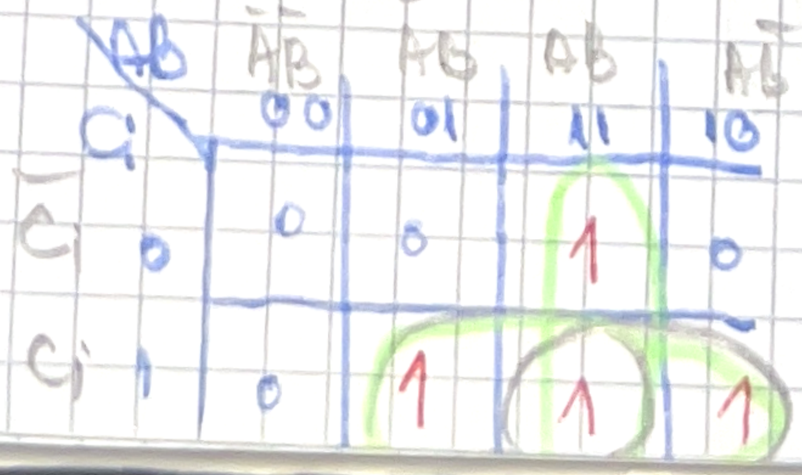
$= \bar{C}_i(\bar{A}B + A\bar{B}) + C_i(\bar{A}\bar{B} + AB) =$

$= \bar{C}_i(A \oplus B) + C_i\overline{(A \oplus B)} =$

$= C_i \oplus (A \oplus B)$

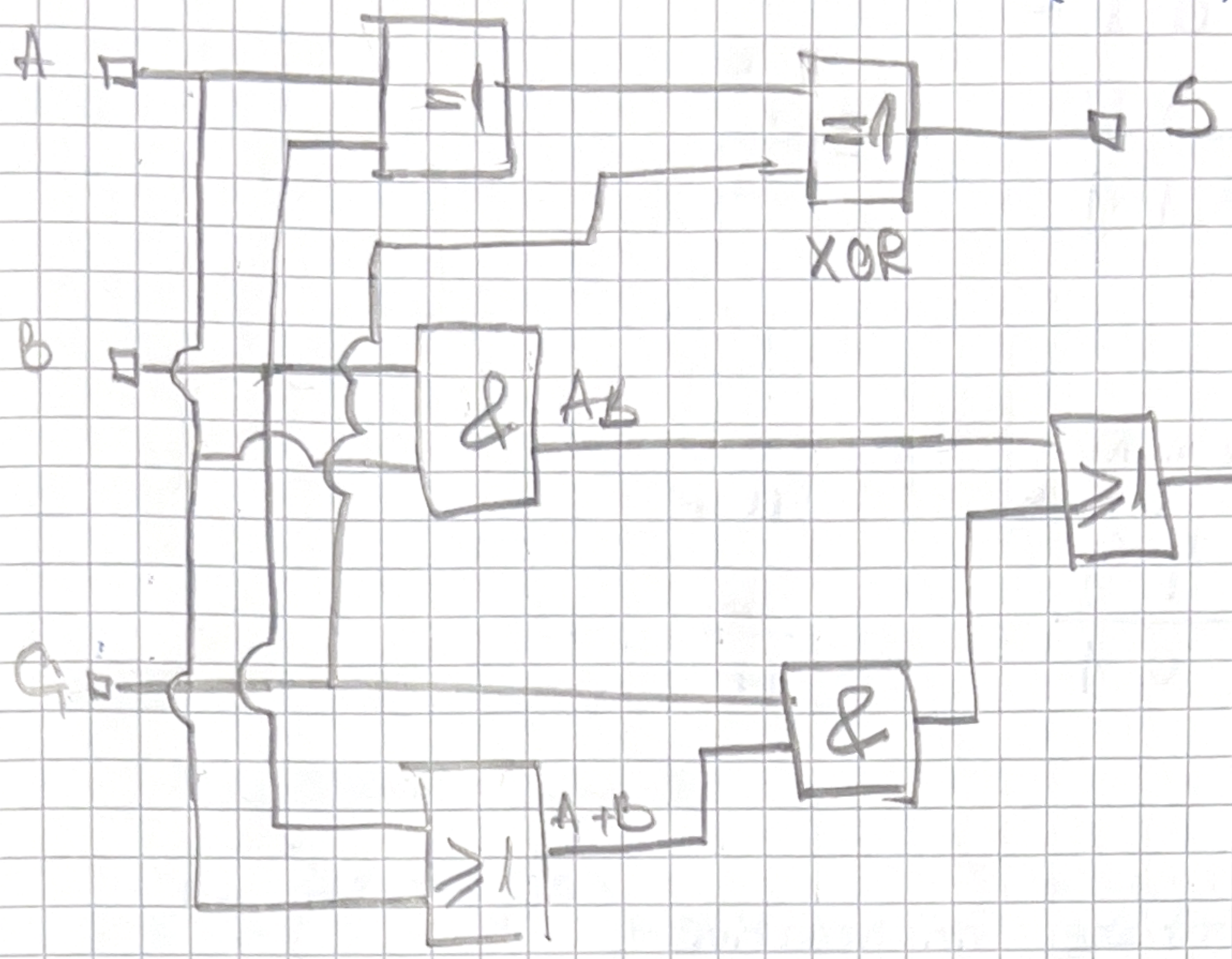
$C_0 = m(3,5,6,7) =$

$= AB + BC_i + AC_i = C_i(A+B)$

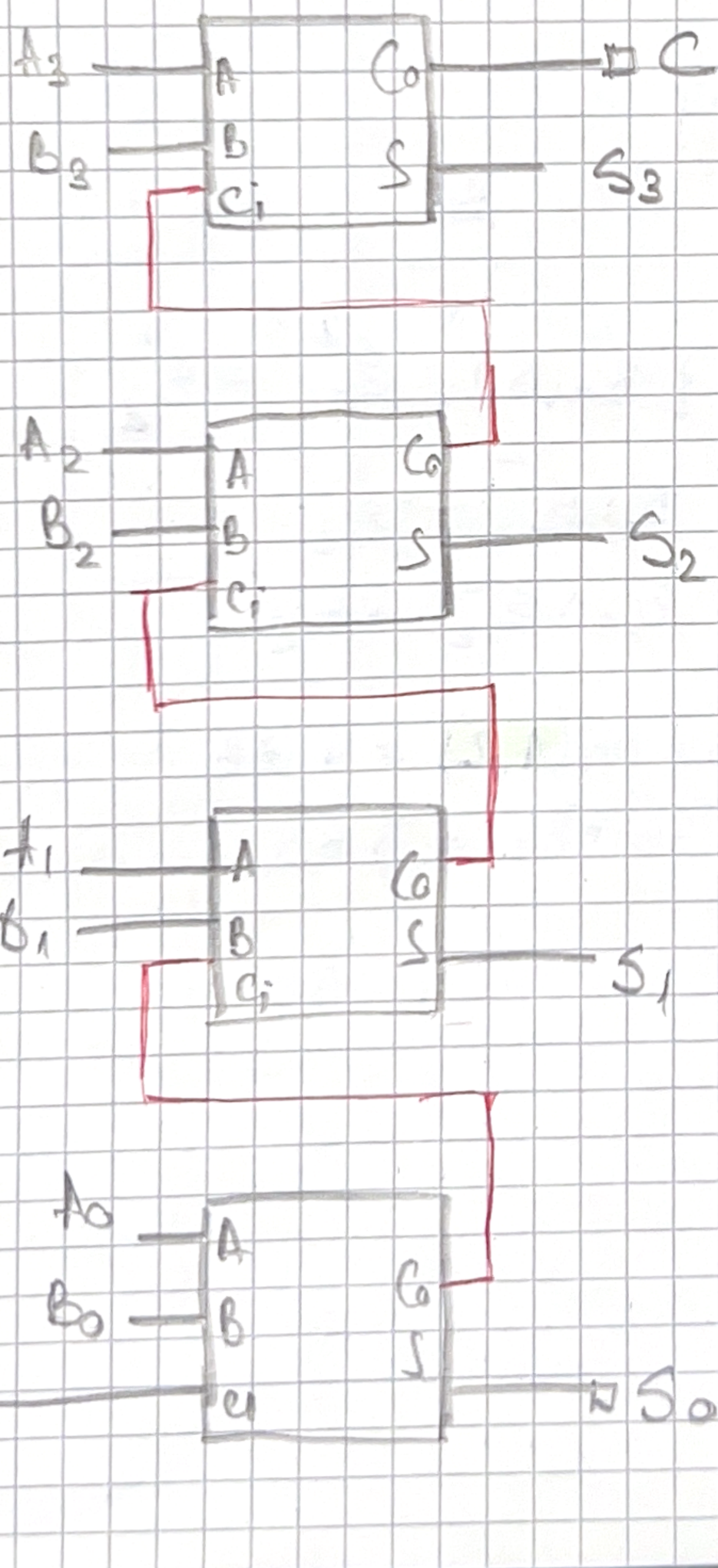


$\bar{A}BC_i \cdot A\bar{B}C_i \cdot AB\bar{C}_i \cdot ABC_i$

$S = C_i \oplus (A \oplus B)$ $C_0 = C_i(A+B)$



SUMATOR PE 4 BITI



MSB — 3 2 1 0 — LSB

A: 0 1 0 1 +

B: 0 1 1 0

1 0 1 1

1

A_3 A_2 A_1 A_0

A: 1 0 1 1 +

B_3 B_2 B_1 B_0

B: 0 1 1 0

C_0 → 1 0 0 0 1

1 1 1

5 +

6

11

11 +

6

17

Reprezentarea nr. negative $\begin{cases} \text{offset} = 2^{k-1} - 1 \\ \text{complement față de 1} \end{cases}$

Half Subtractor HS

Semisçăzătorul

A	B	D	B_0
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

B_0

$\{B_0, D\} : \Sigma_4^2 \to \Sigma_2^2$

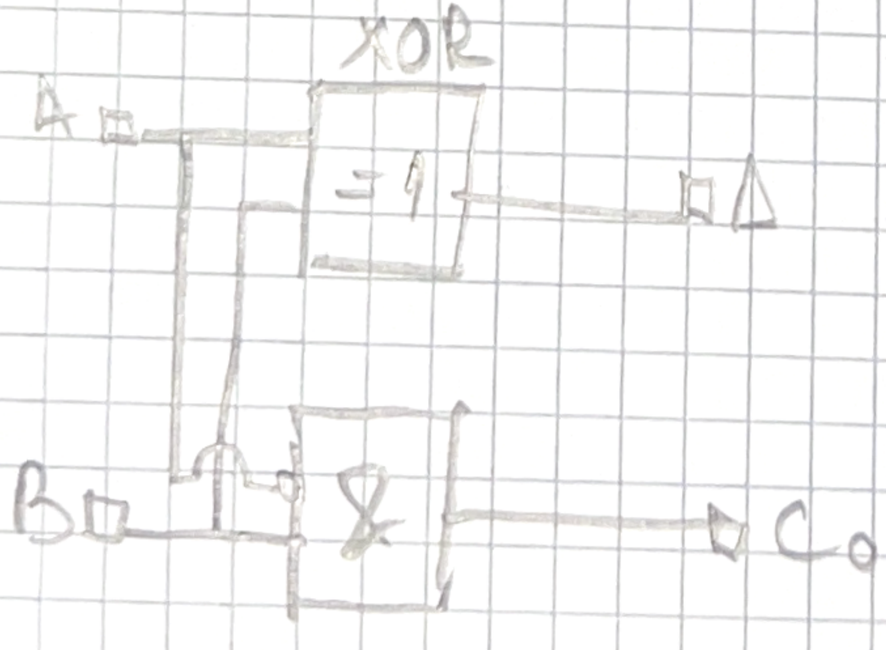
$D : \Sigma_2^2 \to \Sigma_2$

$B_0 : \Sigma_2^2 \to \Sigma_2$

$D = m(1,2) = A \oplus B$

$\bar{A}B + \bar{B}A = A \oplus B$

$C_0 = m(1) = \bar{A}b$



$$101 - 011 = 010$$

$$010$$

$$1$$

$$0010 - 0110 = 1100$$

$$11$$

$$5 - 3 = 2$$

FULL SUBSTRACTOR - FS

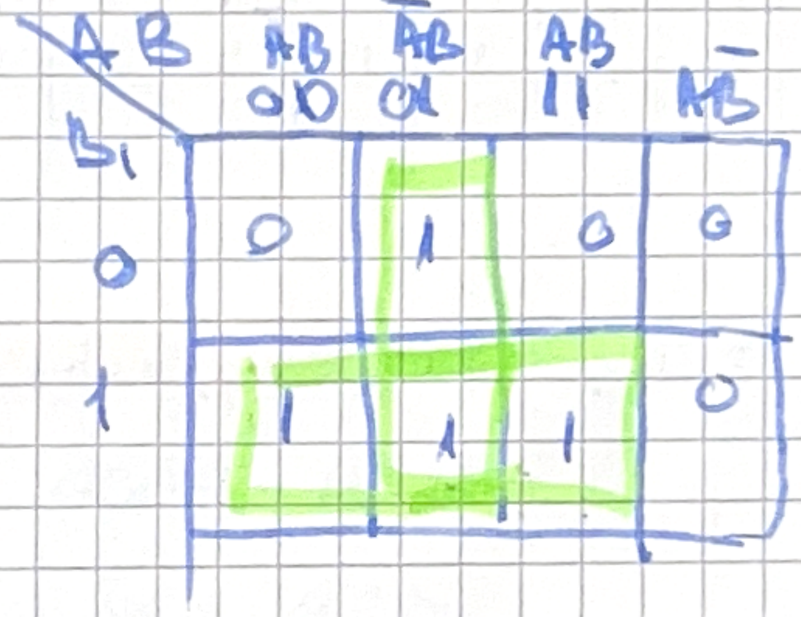
Scăzătorul complet

alg: $A - B = \{B_0, D\}$

A	B	B_i	D	B_0
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

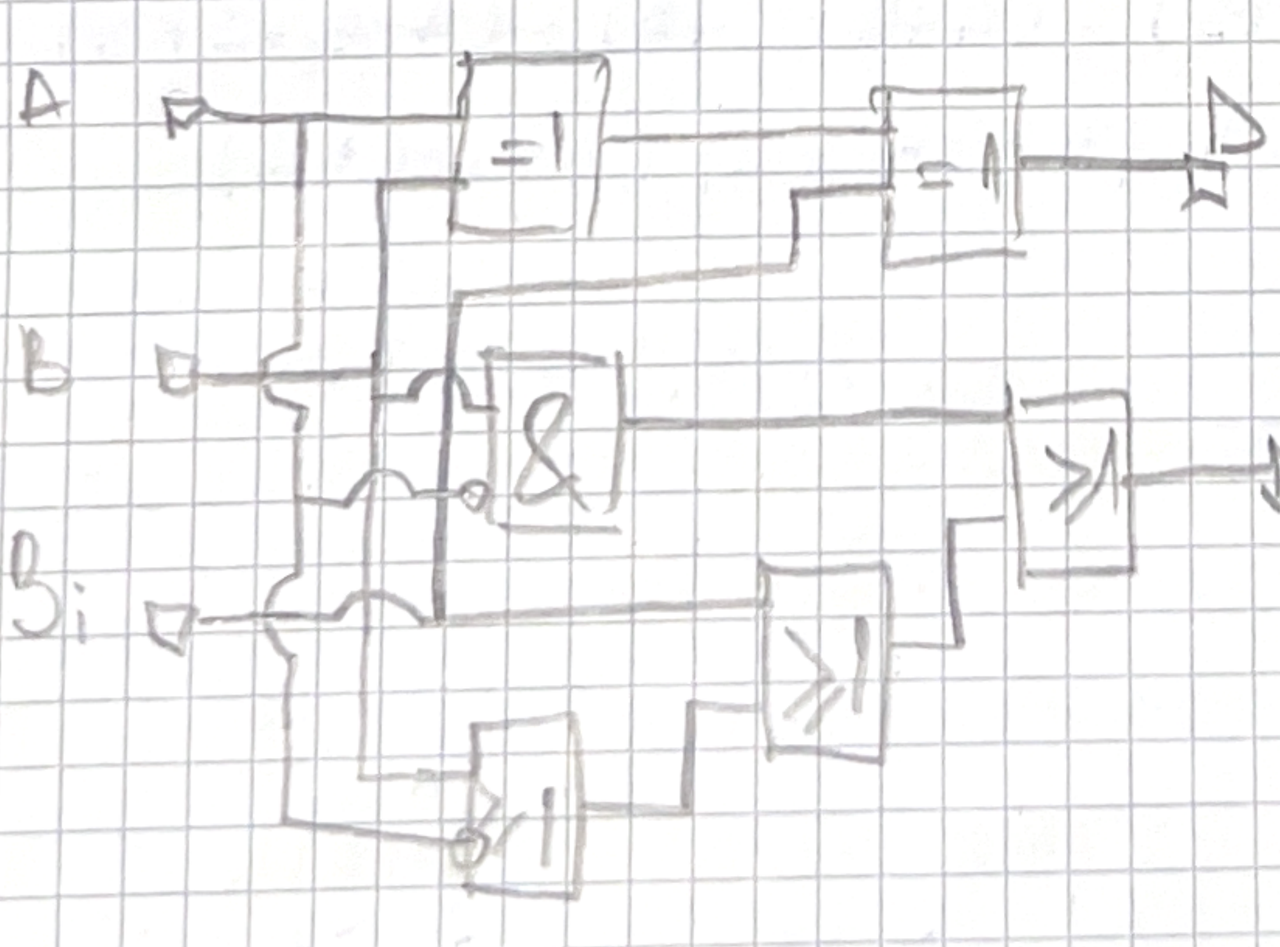
$$D = m(1,2,4,7) = (A \oplus B) \oplus C_i$$

$$B_0 = m(1,2,3,7)$$



$$B_0 = \bar{A}B + \bar{A}C_i + BC_i =$$

$$= \bar{A}B + C_i(\bar{A} + B)$$



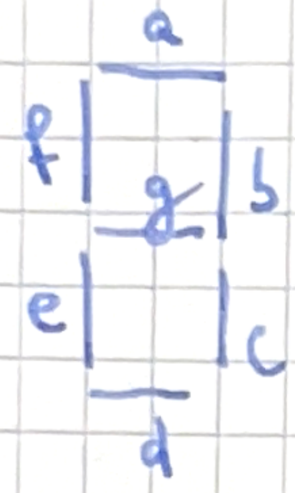
$B_0 = \bar{A}B + B_i(\bar{A} + B)$

$B_{i/o}$ scădere

$C_{i/o}$ - adunare

$C_0 = \bar{A}B + C_i(A + B)$

LP6.



	Intrări A	B	C	D	Ieșiri a	b	c	d	e	f	g
#0	0	0	0	0	1	1	1	1	1	1	0
#1	0	0	0	1	0	1	1	0	0	0	0
#2	0	0	1	0	1	1	0	1	1	0	1
#3	0	0	1	1	1	1	1	1	0	0	1
#4	0	1	0	0	0	1	1	0	0	1	1
#5	0	1	0	1	1	0	1	1	0	1	1
#6	0	1	1	0	1	0	1	1	1	1	1
#7	0	1	1	1	1	1	1	0	0	0	0
#8	1	0	0	0	1	1	1	1	1	1	1
#9	1	0	0	1	1	1	1	1	0	1	1
#A	1	0	1	0	1	1	1	0	1	1	1
#B	1	0	1	1	0	0	1	1	1	1	1
#C	1	1	0	0	0	0	0	1	0	1	1
#D	1	1	0	1	0	1	1	1	0	1	1
#E	1	1	1	0	1	0	0	1	1	1	1
#F	1	1	1	1	1	0	0	0	1	1	1

e) $F_{Cb_e} = m(0,2,6,8,10,11,12,13,14,15) =$

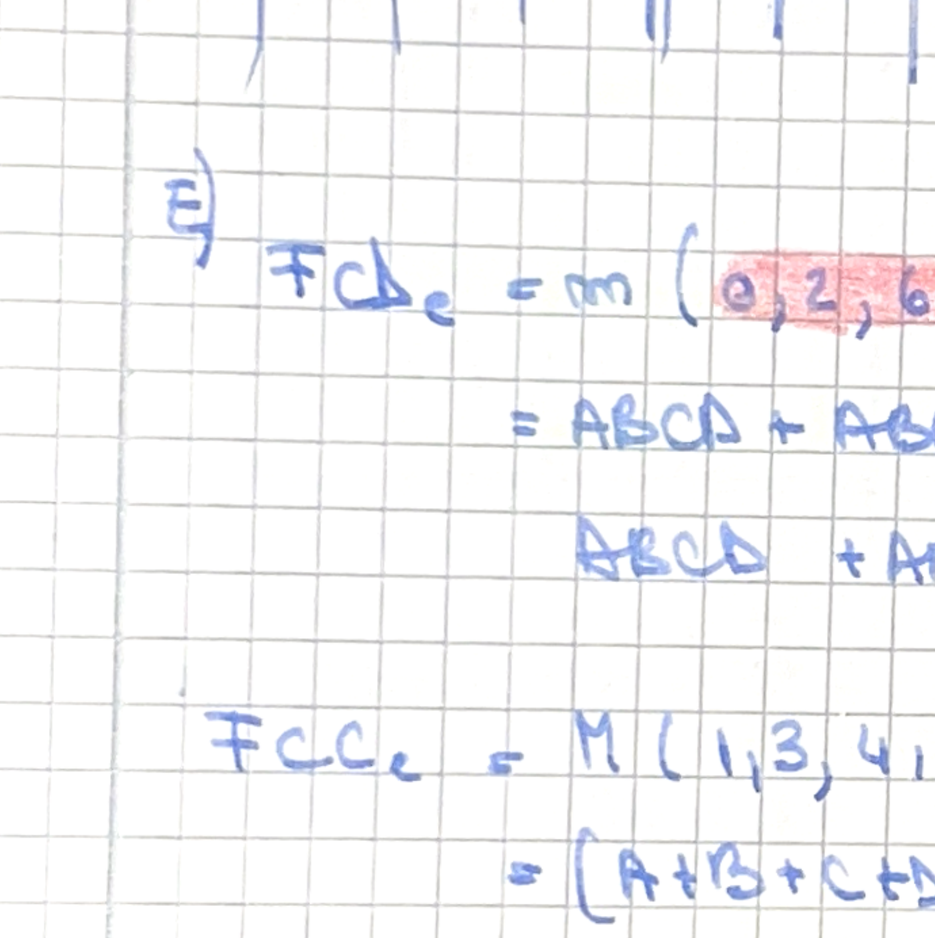
$= \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + \bar{A}BC\bar{D} + A\bar{B}\bar{C}\bar{D} + A\bar{B}C\bar{D} + AB\bar{C}\bar{D} +$

$\bar{A}BCD + A\bar{B}CD + ABCD$

$F_{CC_e} = M(1,3,4,5,7,9)$

$= (A+B+C+D)(A+B+C+\bar{D})(A+B+\bar{C}+D)$

$(A+\bar{B}+C+D)(A+\bar{B}+C+\bar{D})$



- $\bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D} + AB\bar{C}\bar{D} = B\bar{D}$
- $\bar{A}\bar{B}C\bar{D} + \bar{A}BC\bar{D} + ABC\bar{D} + A\bar{B}C\bar{D} = C\bar{D}$
- $AB\bar{C}\bar{D} + ABC\bar{D} + ABCD + AB\bar{C}D = AB$
- $A\bar{B}C\bar{D} + ABC\bar{D} + ABCD + A\bar{B}C\bar{D} = AC$

$e = B\bar{D} + AC + AB + C\bar{D} = \bar{D}(B+C) + A(C+B)$